

Aniruddha Rakshit

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EDUCATION

State University of New York (SUNY) Binghamton

Ph.D. Candidate in Computer Science, Advisor: Jayson Boubin

Binghamton, NY

August, 2022–August, 2027 (expected)

Jahangirnagar University

M.S. in Computer Science and Engineering

Dhaka, Bangladesh

April 2015–February 2017

Jahangirnagar University

B.S. in Computer Science and Engineering

Dhaka, Bangladesh

2010–2014

PUBLICATIONS

- [1] R. Hossain, **Aniruddha Rakshit**, A. Esmaili, B. Edmiston, K.-D. Kang, D. Ismail, and J. Boubin, “WASP: An Edge-AI Framework for Reliable Insect Pest Monitoring in Challenging Sensing and Communication Environments”, in *28th International Conference on Distributed Computing and Networking*, Under Review in ICDCN 2027.
- [2] S. Das, J. Cheng, **Aniruddha Rakshit**, B. Soni, J. Boubin, and R. Ramnath, “Spice: Structured pruning for inference on constrained edge devices”, in *2026 IEEE 23rd Consumer Communications Networking Conference (CCNC)*, 2026, pp. 1–6.
- [3] S. Das, J. Cheng, **Aniruddha Rakshit**, J. Boubin, and R. Ramnath, “Epic: Efficient pruning for inference on constrained devices”, in *Practice and Experience in Advanced Research Computing 2025: The Power of Collaboration*, ser. PEARC '25, Association for Computing Machinery, 2025, ISBN: 9798400713989.
- [4] **Aniruddha Rakshit**, S. Reddy, R. Ramnath, A. Arora, and J. Boubin, “Righteous: Automatic right-sizing for complex edge deployments”, in *2024 IEEE/ACM Symposium on Edge Computing (SEC)*, 2024, pp. 15–28.
- [5] R. Hafiz, M. R. Haque, **Aniruddha Rakshit**, and M. S. Uddin, “Image-based soft drink type classification and dietary assessment system using deep convolutional neural network with transfer learning”, in *Journal of King Saud University - Computer and Information Sciences*, 2022, pp. 1775–1784.
- [6] F. Ahmed, A. R. Choudhury, **Aniruddha Rakshit**, and M. Z. Hasan, “2020 ieee region 10 symposium (tensymp)”, in *An IoT Based System for Printing Braille Letter from Speech*, 2020, pp. 344–347.

WORK EXPERIENCE

State University of New York at Binghamton

Graduate Research Assistant

Binghamton, USA

Spring 2026- Present

State University of New York at Binghamton

Graduate Assistant/ Teaching Assistant

Binghamton, USA

Fall 2022-Fall 2025

Daffodil International University

Assistant Professor

Dhaka, Bangladesh

Spring 2022-Summer 2022

Daffodil International University

Lecturer

Dhaka, Bangladesh

Fall 2015-Spring 2022

PROJECTS

- **FlySpotter (2026-Present)**

- Designed and developed a deployment-aware, context-adaptive SLF detection framework supporting heterogeneous agricultural sensing platforms including camera traps, UAVs, and cloud servers
- Developed a novel optimizer that selects model that satisfies user-defined FPS and accuracy constraints at runtime from a tiered model suite, MobileNetV3, EfficientNet, VGG16, RetinaNet, FasterRCNN, FCOS, YOLO11x/26x, RF-DETR.
- Our optimization algorithm generates 66.1% less sample and uses 86% less time on average than other algorithms.

- **WASP (2025-Present)**

- Designed RPCCL that trains lightweight models to learn domain-resilient representations from limited and corrupted data, mitigating accuracy degradation under domain shift.
- Evaluated on AgriPest-13 and IP102 across 70 physics-based corruptions using MobileNetV3 and ResNet backbones.
- Implemented a location-aware data-rate adaptation(LA-ADR) scheme for LoRaWAN that achieves 77-87% higher data rate and 28-77% higher than standard LoRaWAN ADR.

- **Righteous (2024)**

- Developed Righteous, an edge deployment right-sizing tool for simulators and testbeds to tunable hardware, software, and network parameters.
- Proposed Informed Pareto Simulated Annealing (iPSA) to find near-optimal resource configurations based on user-defined performance goals.
- Achieved up to 3.5× better configurations and 76× faster compared to SOTA techniques.[\[Details\]](#)

- **Spotted Lantern Fly Detection through Drones (2023)**

- Trained YOLOv8 model with datasets from Penn State and the New York State Department of Agriculture and Management
- Submitted at Cornell Institute for Digital Agriculture Hackathon [\[Details\]](#)

- **Robot Task Planning with LLMs (2023)**

- Explored capabilities of LLMs like GPT-3 to interpret grounded instructions and decompose them into robot-executable subtasks. [\[Details\]](#)

SOFTWARE ARTIFACTS

- Deployed our trained model in collaboration with the New York State Department of Agriculture and Markets (NYSDAM).

SCHOLARSHIPS AND AWARDS

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| • ACM/IEEE Symposium on Edge Computing 2024(SEC 2024) Travel Grant | November 2024 |
| • Jahangirnagar University Scholarship (5th Position in class) | December 2013 |
| • Jahangirnagar University Scholarship (7th Position in class) | August 2013 |

SKILLS

- **Programming Languages:** Python, C++, C
- **Machine Learning Libraries:** PyTorch, Tensorflow, Keras, Scikit-Learn, Pandas, Matplotlib
- **Simulation and Design Tools:** PyBullet, MATLAB, ROS
- **System Design Skill:** Docker, Kubernetes, Minikube, Microsoft Azure
- **Embedded Device:** Raspberry Pi, NVIDIA Jetson Orin Nano and Xavier, ESP32